PRAKTIKUM BASIS DATA

MODUL 3

TUGAS PENDAHULUAN

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

TP

1. **Create Table mahasiswa**

```
CREATE TABLE mahasiswa
  2  (id NUMBER PRIMARY KEY, nama VARCHAR2(30),
  3  tempat_lahir VARCHAR2(50),
  4  tanggal_lahir DATE,
  5  nomor_hp VARCHAR2(15),
  6  email VARCHAR2(50),
  7  tinggi_badan NUMBER,
  8  berat_badan NUMBER);
```

Hasil:

```
SQL> desc mahasiswa;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ID                                        NOT NULL NUMBER
 NAMA                                               VARCHAR2(30)
 TEMPAT_LAHIR                                       VARCHAR2(50)
 TANGGAL_LAHIR                                      DATE
 NOMOR_HP                                           VARCHAR2(15)
 EMAIL                                              VARCHAR2(50)
 TINGGI_BADAN                                       NUMBER
 BERAT_BADAN                                        NUMBER
```

2. **Insert Values For table mahasiswa**

INSERT INTO mahasiswa VALUES (1, 'Joko', 'Banyumas', TO_DATE('04-11-2005', 'DD-MM-YYYY'), '081234567', 'joko@gmail.com', 168, 58);
INSERT INTO mahasiswa VALUES (2, 'Dhani', 'Kebumen', TO_DATE('20-05-2006', 'DD-MM-YYYY'), '081234568', 'dhani@gmail.com', 177, 60);
INSERT INTO mahasiswa VALUES (3, 'Dono', 'Cilacap', TO_DATE('12-04-2004', 'DD-MM-YYYY'), '081234569', 'dono@gmail.com', 170, 73);
INSERT INTO mahasiswa VALUES (4, 'Yanti', 'Purbalingga', TO_DATE('23-08-2005', 'DD-MM-YYYY'), '081234565', 'yanti@gmail.com', 165, 53);
INSERT INTO mahasiswa VALUES (5, 'Coki', 'Jakarta', TO_DATE('01-03-2003', 'DD-MM-YYYY'), '081234561', 'coki@gmail.com', 172, 61);
INSERT INTO mahasiswa VALUES (6, 'Prowowo', 'Solo', TO_DATE('07-07-2007', 'DD-MM-YYYY'), '081234562', 'prowowo@gmail.com', 169, 63);
INSERT INTO mahasiswa VALUES (7, 'Habibi', 'Klaten', TO_DATE('19-12-2002', 'DD-MM-YYYY'), '081234563', 'habibi@gmail.com', 178, 80);
INSERT INTO mahasiswa VALUES (8, 'Khofidinia', 'Kebumen', TO_DATE('01-03-2006', 'DD-MM-YYYY'),'081234564','khofidinia@gmail.com',160, 51);
INSERT INTO mahasiswa VALUES (9, 'Marquez', 'Inggris', TO_DATE('01-03-2000', 'DD-MM-YYYY'), '081234565', 'marquez@gmail.com', 162, 66);

INSERT INTO mahasiswa VALUES (10, 'Daniel', 'Amerika', TO_DATE('10-05-2003', 'DD-MM-YYYY'), '081234566', 'daniel@gmail.com', 180, 72);
INSERT INTO mahasiswa VALUES (11, 'Alya', 'Banjarnegara', TO_DATE('26-09-2004', 'DD-MM-YYYY'), '081234518', 'alya@gmail.com', 154, 49);
INSERT INTO mahasiswa VALUES (12, 'Wowo', 'Tangerang', TO_DATE('17-12-2004', 'DD-MM-YYYY'), '081234514', 'wowo@gmail.com', 165, 65);
INSERT INTO mahasiswa VALUES (13, 'Leo', 'Medan', TO_DATE('16-11-2003', 'DD-MM-YYYY'), '081234574', 'leo@gmail.com', 166, 63);
INSERT INTO mahasiswa VALUES (14, 'Yono', 'Palembang', TO_DATE('12-02-2001', 'DD-MM-YYYY'), '081234591', 'yono@gmail.com', 166, 55);
INSERT INTO mahasiswa VALUES (15, 'Yatno', 'Jogjakarta', TO_DATE('11-07-2001', 'DD-MM-YYYY'), '081234531', 'Yatno@gmail.com', 188, 77);
Hasil:

```
SQL> select * from mahasiswa;

        ID NAME
---------- ------------------------------
TEMPAT_LAHIR                       TANGGAL_L NOMOR_HP
-------------------------------------------------- --------- ---------------
EMAIL                              TINGGI_BADAN BERAT_BADAN
-------------------------------------------------- ------------ -----------
         1 Joko
Banyumas                            04-NOV-05 081234567
joko@gmail.com                            168          58

         2 Dhani
Kebumen                             20-MAY-06 081234568
dhani@gmail.com                           177          60

        ID NAME
---------- ------------------------------
TEMPAT_LAHIR                       TANGGAL_L NOMOR_HP
-------------------------------------------------- --------- ---------------
EMAIL                              TINGGI_BADAN BERAT_BADAN
-------------------------------------------------- ------------ -----------

         3 Dono
Cilacap                             12-APR-04 081234569
dono@gmail.com                            170          73

         4 Yanti
Purbalingga                         23-AUG-05 081234565

        ID NAME
---------- ------------------------------
TEMPAT_LAHIR                       TANGGAL_L NOMOR_HP
```

--- --- ---
EMAIL TINGGI_BADAN BERAT_BADAN
--- --- ---
yanti@gmail.com 165 53

5 Coki
Jakarta 01-MAR-03 081234561
coki@gmail.com 172 61

6 Prowowo

ID NAMA
--- ---
TEMPAT_LAHIR TANGGAL_L NOMOR_HP
--- --- ---
EMAIL TINGGI_BADAN BERAT_BADAN
--- --- ---
Solo 07-JUL-07 081234562
prowowo@gmail.com 169 63

7 Habibi
Klaten 19-DEC-02 081234563
habibi@gmail.com 178 80

ID NAMA
--- ---
TEMPAT_LAHIR TANGGAL_L NOMOR_HP
--- --- ---
EMAIL TINGGI_BADAN BERAT_BADAN
--- --- ---
8 Khofidinia
Kebumen 01-MAR-06 081234564
khofidinia@gmail.com 160 51

9 Marquez
Inggris 01-MAR-00 081234565
marquez@gmail.com 162 66

ID NAMA
--- ---
TEMPAT_LAHIR TANGGAL_L NOMOR_HP
--- --- ---
EMAIL TINGGI_BADAN BERAT_BADAN
--- --- ---

```
      10 Daniel
Amerika                                10-MAY-03 081234566
daniel@gmail.com                                180         72
```

```
      11 Alya
Banjarnegara                            26-SEP-04 081234518

      ID NAMA
---------- ------------------------------
TEMPAT_LAHIR                            TANGGAL_L NOMOR_HP
-------------------------------------------------- --------- ---------------
EMAIL                                   TINGGI_BADAN BERAT_BADAN
-------------------------------------------------- ------------ -----------
alya@gmail.com                                  154         49
```

```
      12 Wowo
Tangerang                               17-DEC-04 081234514
wowo@gmail.com                                  165         65
```

```
      13 Leo

      ID NAMA
---------- ------------------------------
TEMPAT_LAHIR                            TANGGAL_L NOMOR_HP
-------------------------------------------------- --------- ---------------
EMAIL                                   TINGGI_BADAN BERAT_BADAN
-------------------------------------------------- ------------ -----------
Medan                                   16-NOV-03 081234574
leo@gmail.com                                   166         63
```

```
      14 Yono
Palembang                               12-FEB-01 081234591
yono@gmail.com                                  166         55

      ID NAMA
---------- ------------------------------
TEMPAT_LAHIR                            TANGGAL_L NOMOR_HP
-------------------------------------------------- --------- ---------------
EMAIL                                   TINGGI_BADAN BERAT_BADAN
-------------------------------------------------- ------------ -----------
      15 Yatno
Jogjakarta                              11-JUL-01 081234531
Yatno@gmail.com                                 188         77
```

15 rows selected.

3. **SELECT Lowercase and Uppercase**

```
SELECT nama,
  2  LOWER(nama) AS lowercase,
  3  UPPER(nama) AS uppercase
  4  FROM mahasiswa;
```

Hasil:

```
NAMA                           LOWERCASE
------------------------------ ------------------------------
UPPERCASE
------------------------------
Joko                           joko
JOKO
                               
Dhani                          dhani
DHANI
                               
Dono                           dono
DONO
                               

NAMA                           LOWERCASE
------------------------------ ------------------------------
UPPERCASE
------------------------------
Yanti                          yanti
YANTI
                               
Coki                           coki
COKI
                               
Prowowo                        prowowo
PROWOWO
                               

NAMA                           LOWERCASE
------------------------------ ------------------------------
UPPERCASE
```

Habibi habibi
HABIBI

Khofidinia khofidinia
KHOFIDINIA

Marquez marquez
MARQUEZ

NAMA LOWERCASE
--- ---
UPPERCASE

Daniel daniel
DANIEL

Alya alya
ALYA

Wowo wowo
WOWO

NAMA LOWERCASE
--- ---
UPPERCASE

Leo leo
LEO

Yono yono
YONO

Yatno yatno
YATNO

15 rows selected.

4. **SELECT INITCAP**

```
SELECT nama,
  2  INITCAP(nama) AS capitalized,
  3  FROM mahasiswa;
```

Hasil:

```
NAMA                           CAPITALIZED
------------------------------ ------------------------------
Joko                           Joko
Dhani                          Dhani
Dono                           Dono
Yanti                          Yanti
Coki                           Coki
Prowowo                        Prowowo
Habibi                         Habibi
Khofidinia                     Khofidinia
Marquez                        Marquez
Daniel                         Daniel
Alya                           Alya
Wowo                           Wowo
Leo                            Leo
Yono                           Yono
Yatno                          Yatno
```

5. **SELECT CONCAT**

```
SELECT nama
  2  ||' - '|| email AS Concat FROM mahasiswa;
```

Hasil:

```
CONCAT
--------------------------------------------------------------------------------
Joko - joko@gmail.com
Dhani - dhani@gmail.com
Dono - dono@gmail.com
Yanti - yanti@gmail.com
Coki - coki@gmail.com
Prowowo - prowowo@gmail.com
Habibi - habibi@gmail.com
Khofidinia - khofidinia@gmail.com
```

Marquez - marquez@gmail.com
Daniel - daniel@gmail.com
Alya - alya@gmail.com
Wowo - wowo@gmail.com
Leo - leo@gmail.com
Yono - yono@gmail.com
Yatno - Yatno@gmail.com

6. **SELECT LENGTH**

SELECT nama,
  2 LENGTH(nama) AS jumlah_huruf FROM mahasiswa;

Hasil:

```
NAMA                           JUMLAH_HURUF
------------------------------ ------------
Joko                                      4
Dhani                                     5
Dono                                      4
Yanti                                     5
Coki                                      4
Prowowo                                   7
Habibi                                    6
Khofidinia                               10
Marquez                                   7
Daniel                                    6
Alya                                      4
Wowo                                      4
Leo                                       3
Yono                                      4
Yatno                                     5
```

7. **SELECT SUBSTR**

SELECT nama,
  2 SUBSTR(nama, 1, 3)AS nama_singkatan
  3 FROM mahasiswa;

Hasil:

```
NAMA                           NAMA_SINGKAT
------------------------------ ------------
Joko                           Jok
Dhani                          Dha
Dono                           Don
```

```
Yanti                    Yan
Coki                     Cok
Prowowo                  Pro
Habibi                   Hab
Khofidinia               Kho
Marquez                  Mar
Daniel                   Dan
Alya                     Aly
Wowo                     Wow
Leo                      Leo
Yono                     Yon
Yatno                    Yat
```

8. **LPAD & RPAD**

```
SELECT nama,
  2  LPAD(nama, 10, '-') AS kiri_nama,
  3  RPAD(nama, 10, '*') AS kanan_nama
  4  FROM mahasiswa;
```

Hasil:

```
NAMA                     KIRI_NAMA
---------------------------- ----------------------------------------
KANAN_NAMA
----------------------------------------
Joko                     ------Joko
Joko******
                                                    
Dhani                    -----Dhani
Dhani*****
                                                    
Dono                     ------Dono
Dono******
                                                    
Yanti                    -----Yanti
Yanti*****
                                                    
Coki                     ------Coki
Coki******
                                                    
Prowowo                  ---Prowowo
Prowowo***
```

Habibi ----Habibi
Habibi****

Khofidinia Khofidinia
Khofidinia

Marquez ---Marquez
Marquez***

Daniel ----Daniel
Daniel****

Alya ------Alya
Alya******

Wowo ------Wowo
Wowo******

Leo -------Leo
Leo*******

Yono ------Yono
Yono******

Yatno -----Yatno
Yatno*****

9. **LTRIM**

SELECT LTRIM (' Dhani') AS hasil_LTRIM,
  2 RTRIM('Dhani ') AS hasil_RTRIM FROM DUAL;

Hasil:

HASIL HASIL
----- -----
Dhani Dhani

10. **INSTR**

SELECT email, INSTR(email,'@') AS posisi_at FROM mahasiswa;

Hasil:

```
EMAIL                                                POSISI_AT
---------------------------------------------------- ----------
joko@gmail.com                                                5
dhani@gmail.com                                               6
dono@gmail.com                                                5
yanti@gmail.com                                               6
coki@gmail.com                                                5
prowowo@gmail.com                                             8
habibi@gmail.com                                              7
khofidinia@gmail.com                                         11
marquez@gmail.com                                             8
daniel@gmail.com                                              7
alya@gmail.com                                                5

EMAIL                                                POSISI_AT
---------------------------------------------------- ----------
wowo@gmail.com                                                5
leo@gmail.com                                                 4
yono@gmail.com                                                5
Yatno@gmail.com                                               6
```

11. REPLACE

SELECT email, REPLACE (email, 'gmail.com', 'telkomuniversity.ac.id') AS new_email FROM mahasiswa;

Hasil:

EMAIL

NEW_EMAIL

---

joko@gmail.com
joko@telkomuniversity.ac.id

dhani@gmail.com
dhani@telkomuniversity.ac.id

dono@gmail.com
dono@telkomuniversity.ac.id

EMAIL

NEW_EMAIL
---
yanti@gmail.com
yanti@telkomuniversity.ac.id

coki@gmail.com
coki@telkomuniversity.ac.id

prowowo@gmail.com
prowowo@telkomuniversity.ac.id

habibi@gmail.com
habibi@telkomuniversity.ac.id

khofidinia@gmail.com
khofidinia@telkomuniversity.ac.id

marquez@gmail.com
marquez@telkomuniversity.ac.id

daniel@gmail.com
daniel@telkomuniversity.ac.id

alya@gmail.com
alya@telkomuniversity.ac.id

wowo@gmail.com
wowo@telkomuniversity.ac.id

leo@gmail.com
leo@telkomuniversity.ac.id

yono@gmail.com
yono@telkomuniversity.ac.id

Yatno@gmail.com
Yatno@telkomuniversity.ac.id

12. MM-MON-MONTH, DD/DR, YYYY-YR-YY

SELECT tanggal_lahir,
  2 TO_CHAR(tanggal_lahir, 'MM-MON-MONTH') FROM mahasiswa;

Hasil:

```
TANGGAL_L TO_CHAR(TANGGAL_LAHIR,'MM-MON-MONTH')
--------- ---------------------------------------
04-NOV-05 11-NOV-NOVEMBER
20-MAY-06 05-MAY-MAY
12-APR-04 04-APR-APRIL
23-AUG-05 08-AUG-AUGUST
01-MAR-03 03-MAR-MARCH
07-JUL-07 07-JUL-JULY
19-DEC-02 12-DEC-DECEMBER
01-MAR-06 03-MAR-MARCH
01-MAR-00 03-MAR-MARCH
10-MAY-03 05-MAY-MAY
26-SEP-04 09-SEP-SEPTEMBER

TANGGAL_L TO_CHAR(TANGGAL_LAHIR,'MM-MON-MONTH')
--------- ---------------------------------------
17-DEC-04 12-DEC-DECEMBER
16-NOV-03 11-NOV-NOVEMBER
12-FEB-01 02-FEB-FEBRUARY
11-JUL-01 07-JUL-JULY
```

SELECT tanggal_lahir,
  2 TO_CHAR(tanggal_lahir, 'DD-DY') AS hari FROM mahasiswa;

Hasil:

```
TANGGAL_L HARI
--------- --------------
04-NOV-05 04-FRI
20-MAY-06 20-SAT
12-APR-04 12-MON
23-AUG-05 23-TUE
01-MAR-03 01-SAT
07-JUL-07 07-SAT
19-DEC-02 19-THU
01-MAR-06 01-WED
01-MAR-00 01-WED
10-MAY-03 10-SAT
26-SEP-04 26-SUN

TANGGAL_L HARI
--------- --------------
17-DEC-04 17-FRI
16-NOV-03 16-SUN
12-FEB-01 12-MON
11-JUL-01 11-WED
```

SELECT tanggal_lahir,
  2 TO_CHAR(tanggal_lahir, 'RR-YY-YYYY') AS Tahun FROM mahasiswa;

Hasil:

```
TANGGAL_L TAHUN
--------- ----------
04-NOV-05 05-05-2005
20-MAY-06 06-06-2006
12-APR-04 04-04-2004
23-AUG-05 05-05-2005
01-MAR-03 03-03-2003
07-JUL-07 07-07-2007
19-DEC-02 02-02-2002
01-MAR-06 06-06-2006
01-MAR-00 00-00-2000
10-MAY-03 03-03-2003
26-SEP-04 04-04-2004

TANGGAL_L TAHUN
--------- ----------
17-DEC-04 04-04-2004
16-NOV-03 03-03-2003
12-FEB-01 01-01-2001
11-JUL-01 01-01-2001
```

13. SYSDATE

SELECT SYSDATE,
  2 TO_CHAR(SYSDATE, 'HH24:MI:SS') AS formatJam_24,
  3 TO_CHAR(SYSDATE, 'HH:MI:SS AM') AS formatJam_12 FROM DUAL;

Hasil:

```
SYSDATE   FORMATJA FORMATJAM_1
--------- -------- -----------
16-MAR-26 01:38:42 01:38:42 AM
```

14. ADD_MONTHS

SELECT NAMA, tanggal_lahir,
  2 ADD_MONTHS(tanggal_lahir, 4) AS empatbulanSetelahLahir FROM mahasiswa;

Hasil:

```
NAMA                           TANGGAL_L EMPATBULA
------------------------------ --------- ---------
Joko                           04-NOV-05 04-MAR-06
Dhani                          20-MAY-06 20-SEP-06
Dono                           12-APR-04 12-AUG-04
Yanti                          23-AUG-05 23-DEC-05
Coki                           01-MAR-03 01-JUL-03
Prowowo                        07-JUL-07 07-NOV-07
Habibi                         19-DEC-02 19-APR-03
Khofidinia                     01-MAR-06 01-JUL-06
Marquez                        01-MAR-00 01-JUL-00
Daniel                         10-MAY-03 10-SEP-03
Alya                           26-SEP-04 26-JAN-05
Wowo                           17-DEC-04 17-APR-05
Leo                            16-NOV-03 16-MAR-04
Yono                           12-FEB-01 12-JUN-01
Yatno                          11-JUL-01 11-NOV-01
```

15. LAST_DAYS

SELECT nama, tanggal_lahir,
  2 LAST_DAY(tanggal_lahir) AS tanggalTerakhirBulan FROM mahasiswa;

Hasil:

```
NAMA                           TANGGAL_L TANGGALTE
------------------------------ --------- ---------
Joko                           04-NOV-05 30-NOV-05
Dhani                          20-MAY-06 31-MAY-06
Dono                           12-APR-04 30-APR-04
Yanti                          23-AUG-05 31-AUG-05
Coki                           01-MAR-03 31-MAR-03
Prowowo                        07-JUL-07 31-JUL-07
Habibi                         19-DEC-02 31-DEC-02
Khofidinia                     01-MAR-06 31-MAR-06
Marquez                        01-MAR-00 31-MAR-00
Daniel                         10-MAY-03 31-MAY-03
Alya                           26-SEP-04 30-SEP-04
Wowo                           17-DEC-04 31-DEC-04
Leo                            16-NOV-03 30-NOV-03
Yono                           12-FEB-01 28-FEB-01
Yatno                          11-JUL-01 31-JUL-01
```

16. NEXT_DAYS

SELECT nama, tanggal_lahir,
  2 NEXT_DAY(tanggal_lahir, 'Sunday') AS mingguBerikutnya FROM mahasiswa;

Hasil:

```
NAMA                           TANGGAL_L MINGGUBER
------------------------------ --------- ---------
Joko                           04-NOV-05 06-NOV-05
Dhani                          20-MAY-06 21-MAY-06
Dono                           12-APR-04 18-APR-04
Yanti                          23-AUG-05 28-AUG-05
Coki                           01-MAR-03 02-MAR-03
Prowowo                        07-JUL-07 08-JUL-07
Habibi                         19-DEC-02 22-DEC-02
Khofidinia                     01-MAR-06 05-MAR-06
Marquez                        01-MAR-00 05-MAR-00
Daniel                         10-MAY-03 11-MAY-03
Alya                           26-SEP-04 03-OCT-04
Wowo                           17-DEC-04 19-DEC-04
Leo                            16-NOV-03 23-NOV-03
Yono                           12-FEB-01 18-FEB-01
Yatno                          11-JUL-01 15-JUL-01
```

17. MONTHS_BETWEEN

SELECT nama,
  2 MONTHS_BETWEEN(SYSDATE, tanggal_lahir) AS selisih_bulan FROM mahasiswa;

Hasil:

```
NAMA                           SELISIH_BULAN
------------------------------ -------------
Joko                                244.38948
Dhani                             237.873351
Dono                              263.131415
Yanti                             246.776577
Coki                              276.486254
Prowowo                           224.292706
Habibi                            278.905609
Khofidinia                        240.486254
Marquez                           312.486254
Daniel                            274.195932
Alya                              257.679802
Wowo                              254.970125
Leo                                      268
Yono                              301.131415
Yatno                             296.163673
```

18. ROUND

SELECT nama, tanggal_lahir,
2 ROUND(tanggal_lahir, 'MONTH') AS pembulatan_bulan,
3 ROUND(tanggal_lahir, 'YEAR') AS pembulatan_tahun FROM mahasiswa;

Hasil:

```
NAMA                           TANGGAL_L PEMBULATA PEMBULATA
------------------------------ --------- --------- ---------
Joko                           04-NOV-05 01-NOV-05 01-JAN-06
Dhani                          20-MAY-06 01-JUN-06 01-JAN-06
Dono                           12-APR-04 01-APR-04 01-JAN-04
Yanti                          23-AUG-05 01-SEP-05 01-JAN-06
Coki                           01-MAR-03 01-MAR-03 01-JAN-03
Prowowo                        07-JUL-07 01-JUL-07 01-JAN-08
Habibi                         19-DEC-02 01-JAN-03 01-JAN-03
Khofidinia                     01-MAR-06 01-MAR-06 01-JAN-06
Marquez                        01-MAR-00 01-MAR-00 01-JAN-00
Daniel                         10-MAY-03 01-MAY-03 01-JAN-03
Alya                           26-SEP-04 01-OCT-04 01-JAN-05
Wowo                           17-DEC-04 01-JAN-05 01-JAN-05
Leo                            16-NOV-03 01-DEC-03 01-JAN-04
Yono                           12-FEB-01 01-FEB-01 01-JAN-01
Yatno                          11-JUL-01 01-JUL-01 01-JAN-02
```